

108. Let the two fractions be a and b . Then, $ab = \frac{14}{15}$ and $\frac{a}{b} = \frac{35}{24}$.

$$\frac{ab}{(a/b)} = \left(\frac{14}{15} \times \frac{24}{35} \right) \Leftrightarrow b^2 = \frac{16}{25} \Leftrightarrow b = \frac{4}{5};$$

$$ab = \frac{14}{15} \Rightarrow a = \left(\frac{14}{15} \times \frac{5}{4} \right) = \frac{7}{6}.$$

Since $a > b$, so greater fraction is $\frac{7}{6}$.

109. $A = 2B \Rightarrow B = \frac{1}{2}A$. So, $AB = \frac{2}{25}$
 $\Rightarrow \frac{1}{2}A^2 = \frac{2}{25} \Rightarrow A^2 = \frac{4}{25} \Rightarrow A = \frac{2}{5}$.

110. Let the fraction be $\frac{a}{1}$.

$$\text{Then, } \frac{1}{a} - a = \frac{9}{20} \Leftrightarrow \frac{1-a^2}{a} = \frac{9}{20}$$

$$\Leftrightarrow 20 - 20a^2 = 9a$$

$$\Leftrightarrow 20a^2 + 9a - 20 = 0$$

$$\Leftrightarrow 20a^2 + 25a - 16a - 20 = 0$$

$$\Leftrightarrow 5a(4a+5) - 4(4a+5) = 0$$

$$\Leftrightarrow (4a+5)(5a-4) = 0 \Leftrightarrow a = \frac{4}{5}. \quad \left[\because a \neq -\frac{5}{4} \right]$$

111. Let the fraction be $\frac{x}{y}$.

$$\text{Then, } x + y = 11 \quad \dots(i)$$

$$\frac{x+1}{y-2} = \frac{2}{3} \Rightarrow 3(x+1)$$

$$= 2(y-2) \Rightarrow 3x - 2y = -7 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 3$ and $y = 8$.

So, the fraction is $\frac{3}{8}$.

112. Let the numerator be x . Then, denominator = $x + 3$.

$$\text{Now, } \frac{x+4}{(x+3)+4} = \frac{4}{5} \Leftrightarrow 5(x+4) = 4(x+7) \Leftrightarrow x = 8.$$

So, the fraction is $\frac{8}{11}$.

113. Let the denominator be x .

Then, numerator = $x + 5$.

$$\text{Now, } \frac{x+5}{x} - \frac{x+5}{x+5} = \frac{5}{4} \Leftrightarrow \frac{x+5}{x} = \frac{5}{4} + 1 = \frac{9}{4} = 2\frac{1}{4}.$$

So, the fraction is $2\frac{1}{4}$.

114. Let the fraction be $\frac{2x}{3x}$.

$$\begin{aligned} \text{Then, } \frac{2x-6}{3x} &= \frac{2}{3} \times \frac{2x}{3x} \Leftrightarrow \frac{2x-6}{3x} \\ &= \frac{4x}{9x} \Leftrightarrow 18x^2 - 54x = 12x^2 \end{aligned}$$

$$\Leftrightarrow 6x^2 = 54x \Leftrightarrow x = 9.$$

Hence, numerator of the original fraction = $2x = 18$.

115. Let the fraction be $\frac{x}{y}$. Then,

$$\frac{x}{y+1} = \frac{1}{2} \Leftrightarrow 2x - y = 1 \quad \dots(i)$$

$$\text{and, } \frac{x+1}{y} = 1 \Leftrightarrow x - y = -1 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 2, y = 3$.

Hence, the required fraction is $\frac{2}{3}$.

116. Let the fraction be $\frac{x}{y}$. Then,

$$\frac{x+2}{y+3} = \frac{7}{9} \Leftrightarrow 9x - 7y = 3 \quad \dots(i)$$

$$\text{and } \frac{x-1}{y-1} = \frac{4}{5} \Leftrightarrow 5x - 4y = 1 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 5, y = 6$.

Hence, the original fraction is $\frac{5}{6}$.

117. Let the fraction be $\frac{x}{y}$. Then,

$$\frac{x+\frac{1}{4}}{y-\frac{1}{3}} = \frac{33}{64} \Leftrightarrow \frac{3(4x+1)}{4(3y-1)} = \frac{33}{64} \Leftrightarrow \frac{4x+1}{3y-1} = \frac{33}{64} \times \frac{4}{3} = \frac{11}{16}$$

$$\Leftrightarrow 16(4x+1) = 11(3y-1)$$

$$\Leftrightarrow 64x + 16 = 33y - 11$$

$$\Leftrightarrow 64x - 33y = -27, \text{ which cannot be solved to find } \frac{x}{y}.$$

Hence, the original fraction cannot be determined from the given data.

118. Let the fraction be $\frac{x}{y}$.

$$\text{Then, } \frac{x+200\% \text{ of } x}{y+300\% \text{ of } y} = \frac{15}{26} \Leftrightarrow \frac{3x}{4y} = \frac{15}{26} \Leftrightarrow \frac{x}{y} = \frac{15}{26} \times \frac{4}{3} = \frac{10}{13}.$$

119. Let the fraction be $\frac{x}{y}$.

$$\text{Then, } \frac{x+4}{y} - \frac{x}{y} = \frac{2}{3} \Leftrightarrow \frac{4}{y} = \frac{2}{3} \Leftrightarrow y = \left(\frac{4 \times 3}{2} \right) = 6.$$

$\therefore$ Denominator = 6.

120. Let the fraction be $\frac{x}{y}$.

$$\text{Then, } \frac{110\% \text{ of } 2x}{70\% \text{ of } 3y} = 11\% \text{ of } \frac{16}{21} \Leftrightarrow \frac{22x}{21y} = \frac{11}{100} \times \frac{16}{21}$$

$$\Leftrightarrow \frac{x}{y} = \left(\frac{11}{100} \times \frac{16}{21} \times \frac{21}{22} \right) = \frac{2}{25}.$$

121. Let the two parts be $(54 - x)$ and x .